# A MAP FROM A FINITE MODULE THAT FACTORS THROUGH A PROJECTIVE MODULE FACTORS THROUGH A FINITE PROJECTIVE ONE

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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*Allowed standard prerequisites (editorial).*

- Factoring through a projective module is equivalent to factoring through a free module, and the property is preserved by composition. the statement counted by the sibling card S1-algebra-ideal-factor-through-projective at 17711-17735, cited at proof lines 17745-17747. Supplies the factorisation through a free module that the proof then shrinks.
- A map into a direct sum of copies of the ring has image contained in a finite subsum when the source is finite. used at proof lines 17747-17751. Produces the finite free module through which the map factors.

*Source reference.* Original lemma 0G91: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20286–20286. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

*Source reference.* Original lemma 0G90: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 20285–20285. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

**Lemma 0.2.** *Let $R$ be a ring. Let $\varphi : M \to N$ be an $R$-module map. If $\varphi$ factors through a projective module and $M$ is a finite $R$-module, then $\varphi$ factors through a finite projective module.*

*Date*: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Proof.* By Lemma 0G90 we can factor $\varphi = \tau \circ \sigma$ where the target of $\sigma$ is $\bigoplus_{i \in I} R$ for some set $I$. Choose generators $x_1, \ldots, x_n$ for $M$. Write $\sigma(x_j) = (a_{ji})_{i \in I}$. For each $j$ only a finite number of $a_{ij}$ are nonzero. Hence $\sigma$ has image contained in a finite free $R$-module and we conclude. $\square$